

X-Road Colombia: An Automated Maturity Assessment Platform for Public-Sector Interoperability

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Abstract. This paper presents *X-Road Colombia*, a full-stack platform that automates the diagnosis of digital interoperability maturity in the Colombian public sector. Colombia adopted X-Road as the backbone of its interoperability layer and defined, through MinTIC, a framework structured in four domains: legal, organizational, semantic, and technical. No independent instrument currently contrasts administrative registration with verifiable operational capability. The platform operationalizes the MinTIC framework into measurable indicators computed over a registry of entities, their published services, and their declared X-Road state, combining a deterministic maturity engine, HTTP-based API introspection, semantic validation, gap analysis, clustering and prediction models, geospatial and network visualizations, and multi-format reporting behind a hardened REST API. The diagnostic over 127 registered entities yields a national maturity index of 2.3/5.0 ($SD = 1.1$), a *Basic* level, with the technical domain leading (3.1) and the semantic domain weakest (1.9). The study exposes a registration–operation gap: 32% of the registered entities remain at the initial level, and the Hacienda y Finanzas sector concentrates the highest maturity (3.8). An automatic cross-check reproduces the same domain ordering (semantic 45.3, organizational 49.3, legal 55.7, technical 58.3 on a 0–100 scale). Results indicate that evidence-based automated auditing is feasible and that connectivity alone does not produce interoperability.

Keywords: Interoperability · X-Road · Digital government · Maturity model · Automated assessment · Colombia

1 Introduction

Colombia has prioritized state modernization as a cross-cutting public policy objective, materialized in a digital transformation agenda and in normative instruments that mandate transparency, open data, and efficient public service delivery [4,2,3]. Within that agenda, digital interoperability is a foundational capability: the organizational and technological condition that allows heterogeneous public institutions to exchange information securely, timely, and with a shared meaning, so that citizens are not asked for data the State already holds.

The country adopted X-Road as the interoperability backbone of the State and defined, through MinTIC, an *Interoperability Framework for Digital Government* organized in four domains—legal, organizational, semantic, and technical [5]. X-Road is a distributed data-exchange layer in which each organization runs a security server that publishes and consumes services through machine-processable contracts and signed messages [7]; its operational inventory is coordinated by the National Digital Agency (AND) [6]. Despite this architecture, implementation has progressed unevenly: some entities expose integrated, documented services while many others remain at early digitalization stages. More importantly, the symptoms of that asymmetry—duplicated data requests, routing errors, undetected service failures—cannot be diagnosed with the information that is publicly available, because administrative registration in a catalog is not the same as operational interoperability.

This work addresses the absence of an *independent, automated, and replicable* instrument that measures entity maturity against the national framework, contrasts registration with verifiable service availability, localizes the domains and sectors where deficits concentrate, and produces traceable evidence for prioritization. Manual reviews and self-declaration instruments are insufficient: they are costly, hard to repeat, and systematically overestimate readiness. The remainder of this paper reviews the conceptual framework (Section 2), the architecture (Section 3), the implementation (Section 4), the evaluation (Section 5), the security and governance model (Section 6), and the conclusions (Section 7).

Contributions. (i) An operational model translating the four MinTIC domains into measurable indicators; (ii) a deterministic maturity engine with an explicit rubric and ordinal levels that removes subjective valuation; (iii) a software instrument that verifies evidence automatically through API introspection, semantic validation, and severity-ranked gap analysis; (iv) a visual analytics layer—geospatial map, service matrix, and relationship graph—that exposes the state interconnection map; and (v) a reproducible engineering blueprint, deployable with one Docker Compose command and hardened with authentication, authorization, and auditing.

Research questions. RQ1: what is the interoperability maturity level of the entities registered in the national X-Road node when measured against the MinTIC framework? RQ2: what gap exists between formal registration and verifiable operational service availability? RQ3: which domains concentrate the largest deficits? RQ4: can an automated assessment instrument improve decision-making compared with administrative self-reporting?

2 Conceptual Framework and Related Work

Interoperability is the capability of heterogeneous organizations to exchange information for mutually beneficial goals [8]. In the public sector it is broader than connectivity: it requires coordinated processes, harmonized legal conditions, and aligned data meaning. The reference frameworks converge on a

four-layer decomposition—legal, organizational, semantic, and technical—which means that two institutions can be physically connected and still be unable to produce a legally valid, semantically correct, and operationally useful exchange. The semantic layer is the most fragile: without shared dictionaries, code lists, and canonical schemas, each pair of entities must negotiate bilateral mappings, which does not scale and silently degrades data quality.

MinTIC’s framework defines the reference architecture, standards, and governance mechanisms for public data exchange [5] and organizes assessment into four domains: *legal* (normative basis and lawful processing), *organizational* (governance, roles, and processes), *semantic* (shared meaning, metadata, and vocabularies), and *technical* (connectivity, security, API quality, and non-functional attributes). On top of the domains it defines a staged maturity progression. The platform operationalizes four computable levels—*Inicial*, *Básico*, *Intermedio*, and *Avanzado* (Section 3.1)—and the diagnostic of Section 5 additionally uses a normalized 1–5 index.

X-Road, maintained by the Nordic Institute for Interoperability Solutions [7], is member-centric: each organization registers with a member class and code and deploys security servers that publish and consume services. Because data stays in the source systems and moves through authenticated, signed, and logged transactions, X-Road became the backbone of several national architectures, including Colombia’s. For diagnostic purposes, however, the member registry is insufficient: being *registered* proves enrollment, not that the entity publishes a documented, reachable, and consumable service.

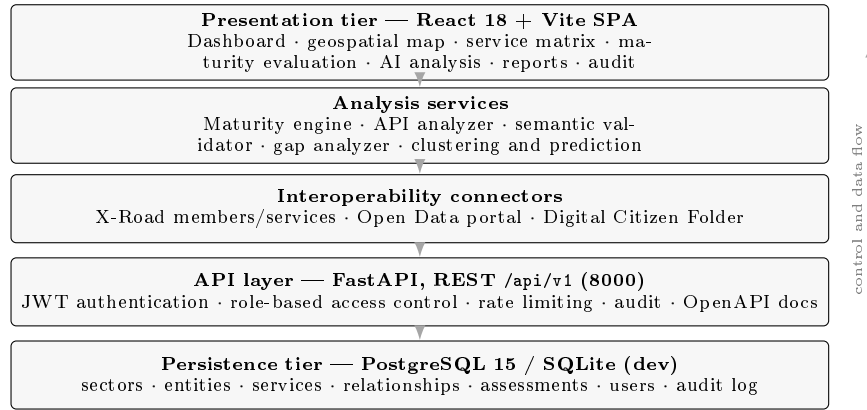
Maturity models structure evolutionary capability paths. De Bruin et al. [9] formalized their design phases and Becker et al. [10] established design principles for IT management maturity models; in electronic government, recent work surveys the landscape of e-government maturity models [11] and addresses cross-organizational data interoperability maturity [12]. Three limitations persist: instruments are usually *self-declared* (upward bias), *manual* (expensive to repeat), and *static* (disconnected from artifacts that change continuously). This work inverts the evidence flow: the platform computes the level from artifacts and observable attributes instead of asking entities to declare it. Consequently, the gap addressed here is the shortage of open, automated, and reproducible instruments that measure interoperability maturity from verifiable evidence in a specific national regulatory context. Table 1 positions the approach against the dominant alternatives.

3 System Architecture

The platform is a decoupled three-tier client–server system (Fig. 1): a single-page application for visualization and interaction, a REST API that concentrates business logic and assessment, and a relational persistence tier, all orchestrated with Docker and communicating exclusively through the HTTP contract. Three design decisions shape it. The *scoring rules are deterministic and explicit*, so every point is attributable to a named indicator and a result can be audited. *Evidence*

Table 1. Comparison of approaches for public-sector interoperability assessment across functional dimensions.

Approach	Evidence based	Auto. scoring	Multi-domain	Service verification	Ecosys. visual.
Administrative self-reporting	No	No	Partial	No	No
Static BI dashboard on registry	Partial	No	No	No	Partial
Manual maturity survey / expert panel	Partial	No	Yes	No	No
Technical monitoring of exchange layer	Yes	Yes	No	Yes	Partial
Proposed platform	Yes	Yes	Yes	Yes	Yes

**Fig. 1.** Layered architecture of the proposed platform. Arrows indicate the control and data flow between tiers.

is stored next to the result, since each assessment keeps its domain scores, criteria, assessor, date, and recommendations, enabling historical comparison. And *verification is separated from visualization*: connectors and analyzers produce evidence, the maturity engine converts it into scores, and the interface consumes aggregated results.

Data model. The schema has seven tables. *Sector* groups entities by domain of public action. *Entity* stores identity (name, acronym, NIT, sector), territorial location (department, municipality, coordinates), contact channels, the X-Road state (**not_connected**, **pending**, **connected**, with member code and connection date), and the CIO contact fields used for fieldwork. *Service* describes what is published: category, protocol (**REST**, **SOAP**, **X-Road**), endpoint URL, API version, declared data, semantic and security standards, status, and documentation URL. *Relationship* models the directed exchange between a source and a target entity with its protocol and state, and is the substrate of the ecosystem graph. *MaturityAssessment* stores, per entity and date, the global level and score, the four domain scores, six detailed criteria, the assessor, and the recommendations. Finally, *Usuario* and *Auditoria* support authentication and traceability.

Table 2. Operationalization of the MinTIC interoperability domains into computable indicators.

Domain	Computable indicator and evidence source	Platform artifact
Technical	X-Road connection state, number and protocol of published services, documented endpoints; evidence from the X-Road registry, the service catalog, and HTTP introspection	<code>technical_domain_score</code> , <code>POST /interop/api-analysis</code>
Semantic	Standardized metadata, declared data and semantic standards, naming and structure conventions; evidence from service records, API responses, and validation rules	<code>semantic_domain_score</code> , <code>POST /interop/validation</code>
Organizational	Sector classification, governance artifacts, trained personnel criterion, CIO contactability; evidence from the entity registry and criteria fields	<code>organizational_domain_score</code> , <code>has_trained_personnel</code>
Legal	Compliance statements, declared security and data standards, availability of institutional channels; evidence from entity and service records	<code>legal_domain_score</code> , <code>has_interoperability_policy</code>

Because the X-Road state is a first-class attribute, the registration–operation gap is computed directly as a query, and because assessments are append-only, the evolution of an entity can be reconstructed.

3.1 The Maturity Assessment Engine

The engine converts attributes and evidence into a composite score $S \in [0, 100]$ and an ordinal level, using an explicit rubric:

$$S = 25 s_{\text{xroad}} + 25 s_{\text{services}} + 20 s_{\text{contact}} + 15 s_{\text{sector}} + 15 s_{\text{geo}} \quad (1)$$

where $s_{\text{xroad}} = 1$ if the entity is connected to X-Road and 0.4 if the connection is pending; $s_{\text{services}} = \min(n/4, 1)$ with n the number of published services; s_{contact} is the weighted presence of website, email, and phone; $s_{\text{sector}} = 1$ if the entity is classified in a sector; and $s_{\text{geo}} = 1$ if valid coordinates are registered. The level follows thresholds: *Inicial* for $S < 25$, *Básico* for $25 \leq S < 50$, *Intermedio* for $50 \leq S < 75$, and *Avanzado* for $S \geq 75$. In parallel, the engine derives the four MinTIC domain scores from the same evidence, which allows the diagnostic to be reported by domain. Table 2 summarizes the operationalization, and the engine is exposed through `/maturity/assessments` with a bulk operation that generates assessments for the active entities that lack one.

Connectors and verification. Evidence acquisition is encapsulated in replaceable services. The *X-Road connector* models members (member code, class, status REGISTERED/APPROVED/DELETION, service count, last activity), published services (code, provider, protocol, state, response time), ecosystem statistics, and the connectivity report. The *Open Data connector* queries the national open data portal for datasets and sector statistics, and the *Digital Citizen Folder client*

consumes the digital citizen service catalog and aggregates usage statistics. The *API analyzer* performs non-intrusive HTTP introspection: it detects the protocol family (SOAP/WSDL, GraphQL, gRPC, REST), finds documentation and OpenAPI/Swagger specifications, counts endpoints, extracts authentication mechanisms, and produces a quality score with issues and recommendations. The *semantic validator* adds rule-based conformance checks in six categories—naming, structure, parameters, response, documentation, and security—verifying conventional operational endpoints (`/health`, `/version`, `/docs`, `/api/v1`) and recommended security headers. The *gap analyzer* converts evidence into prioritized gaps on a five-step severity scale and six categories, with expected standards such as at least three endpoints, mandatory REST availability, documentation completeness of 80%, HTTPS with OAuth2 or API-key authentication, a success rate of 95%, and an average response time below 1.5 s. The external connectors implement the full contract of the integration but currently return simulated payloads, because live access to the national systems requires an institutional agreement; the analyzers, in contrast, operate on real network responses when an endpoint is provided. This asymmetry is treated as a threat to validity in Section 5.2.

Analytics, visualization, and reporting. Two analytical layers sit on the evidence: a descriptive one (aggregations by sector, department, X-Road state, and protocol, plus the maturity distribution) and an inferential one (clustering with an optimal- k search, a maturity predictor trained on historical assessments, a recommendation engine, and a risk and action-plan generator). The presentation tier materializes the results in a KPI dashboard, a geospatial map filtered by maturity and department, a service matrix by sector and protocol, a maturity view with the radar of the four domains, and an audit view; the relationship table is rendered as a force-directed graph, the visual counterpart of the question of which entities actually exchange data. Reporting exports entities, services, and maturity in CSV and XLSX and generates JSON, HTML, and PDF reports, complemented by a client-side PDF export. An embedded assistant answers natural language questions over the live database with a bounded context window.

4 Implementation

The backend is a Python 3.11 service on FastAPI 0.104 with Uvicorn, using SQLAlchemy 2.0 over PostgreSQL 15 in production and SQLite in development, with Pydantic 2.5 validating every contract (for example, rejecting an overall level outside the 1–4 range or a score outside 0–100 before it reaches the database). Analytics use pandas, NumPy, and scikit-learn; server-side reporting uses ReportLab and openpyxl; JWT signing uses `python-jose` and password hashing uses `passlib` with `bcrypt`; endpoint verification uses `httpx`. The frontend is a React 18 single-page application bundled with Vite 5 and styled with Tailwind, with Leaflet for the geospatial view, Recharts for KPI and radar charts, and `react-force-graph-2d` with `d3-force` for the ecosystem graph; it is bilingual (Spanish and English) through `i18next`.

All functionality is exposed under the versioned prefix `/api/v1`, grouped into eleven routers (`auth`, `sectors`, `entities`, `services`, `maturity`, `dashboard`, `reports`, `ai`, `interop`, `relationships`, `agentgd`) covering CRUD operations, bulk assessment generation, KPIs, connector and analyzer operations, the ecosystem graph, and exports, with OpenAPI documentation at `/api/docs`. The lifecycle is deliberately auditable: an administrator maintains entities and services, the engine evaluates a single entity or generates assessments for all active entities that lack one, recording “Sistema Automático” as the assessor for machine-generated results, while write operations require the administrator role and authenticated analysts retain read access.

The whole environment is reproducible with one Docker Compose command orchestrating PostgreSQL 15 (not published on the host), the API on port 8000, and the Nginx-served frontend on port 5173; on startup the backend creates the schema and seeds the initial accounts with bcrypt hashes. Seed scripts load the official entity list, the service catalog, the relationship links, and the domain assessments, and a migration utility transfers a local SQLite database to the managed instance. Production runs on managed free-tier infrastructure—CDN-hosted frontends, containerized backends, and a managed PostgreSQL instance requiring TLS in transit—with every configuration value (signing secret, token lifetimes, rate limit, CORS allowlist, initial credentials) injected through environment variables.

5 Experimental Evaluation

5.1 Setup and Results

The evaluation follows a descriptive–exploratory, mixed-method design with a cross-sectional cut-off date of 20 March 2026. The primary evidence source is the official response to a right-of-petition request filed with the National Digital Agency (record AND-2026-0014789), which reports 124 operating entities and 67 additional entities in the connection phase; secondary sources are the national open data portal and the institutional transparency portals used to confirm the existence and documentation of published services. The population of interest is the set of 127 public entities registered in the central X-Road node; entities that have not started the integration process are explicitly out of scope, which is a deliberate boundary and not a claim about the whole State. Two scales are reported: the *operational composite* $S \in [0, 100]$ of Eq. (1) with its ordinal level, and the *normalized maturity index* $\bar{M} \in [1, 5]$, defined as the mean of the four domain scores and used to state the national diagnosis. Eight scenarios (T1–T8) exercised the instrument end to end: national aggregation, sector and department distribution, bulk maturity generation, domain comparison, registration–operation gap, geospatial localization, endpoint introspection with semantic validation, and gap analysis with report export.

The national maturity index is $\bar{M} = 2.3$ on the 1–5 scale ($SD = 1.1$), placing the ecosystem at the *Basic* level: objectives and first technical processes exist, but use of the interoperability layer is not yet operationally consolidated

for most entities. This answers RQ1. The decomposition by domain is more informative than the aggregate: the technical domain leads with a mean of 3.1, consistent with connectivity being the most visible and best-instrumented part of the process, whereas the semantic domain is the weakest with 1.9 and the organizational and legal domains occupy intermediate positions. Connectivity has advanced faster than the agreements on what the connected data means, which answers RQ3: the deficit concentrates in the semantic layer, with the organizational layer as the second constraint.

The strongest structural result is that registration and operational capability are not equivalent: 32% of the registered entities remain at the initial level, meaning that enrollment is not accompanied by installed technical capability or by a verifiable published service. The gap affects roughly one third of the formally integrated ecosystem, which answers RQ2 and explains why a self-declared compliance view overestimates the readiness of the State. Sector asymmetry compounds the problem: Hacienda y Finanzas concentrates the highest maturity (3.8), sustained by the tax and finance authority and the finance ministry, whose data-exchange needs are recurring and legally mandatory, while sectors with less transactional services and territorial entities with weaker technology teams trail behind.

As an implementation cross-check, the engine of Eq. (1) was run over the seeded registry to generate automatic assessments for the entities that had none. The resulting domain ordering reproduces the national diagnosis—semantic weakest, technical strongest—with composites of 45.3, 49.3, 55.7, and 58.3 respectively and an overall composite of 52.1. The consistency between two independent computations indicates that the instrument does not reproduce a hard-coded conclusion and answers RQ4: compared with self-reporting, it produces a repeatable measurement, an explicit reason for every point, and a prioritized list traceable to evidence. Table 3 consolidates the empirical results.

Runtime behaviour. Two properties of the public deployment are reported as infrastructure observations, not as a controlled benchmark: the container platform suspends idle services after about fifteen minutes, so the first request can take up to roughly fifty seconds, and the API enforces a global limit of 120 requests per minute per client address. Outbound verification is time-bounded (twelve seconds for endpoint introspection and sixty seconds for the assistant), which prevents a slow external service from blocking the request path. No formal latency benchmark was executed.

5.2 Threats to Validity

Internal: the rubric is a proxy, since domain scores derive from registry attributes and observable artifacts rather than from a full audit inside each entity, so a well-classified entity with a complete contact record can obtain points that do not correspond to genuine capability; additionally, connector-derived indicators describe the integration contract rather than the live ecosystem, and endpoint introspection relies on self-exposed documentation with certificate verification

Table 3. Consolidated results of the diagnostic over the registered ecosystem and cross-check of the automated engine (domain values on the 1–5 scale, with the 0–100 composite in parentheses).

Indicator	Value	Reading
National maturity index	2.3/5.0 (SD 1.1)	<i>Basic</i> level: not stalled, far from integrated
Technical domain	3.1 (58.3)	Strongest; connectivity mostly resolved
Semantic domain	1.9 (45.3)	Weakest; the binding constraint
Organizational domain	intermediate (49.3)	Partial governance; territorial entities lag
Legal domain	intermediate (55.7)	Normative base in place, evidence incomplete
Entities at the initial level	32%	Registration did not become installed capability
Leading sector	Hacienda y Finanzas 3.8	Driven by the tax and finance authorities
Overall engine cross-check	52.1	<i>Intermedio</i> threshold; ordering consistent with the national diagnosis

relaxed. *Construct*: operational subtleties such as the real consumption of a service by another entity are only partially modeled through the relationship table, and the four computable levels simplify a finer staged national model, so level labels are operationalized categories and not official classifications. *External*: the scope is the Colombian X-Road context and the entities registered in the central node, the result is a time-bound snapshot that must be re-run to remain valid, and no claim is made about other national architectures. *Conclusion*: the study is descriptive without a control group, so the improvement in decision-making (RQ4) is supported by traceability and the elimination of manual valuation rather than by a randomized trial; the automatic cross-check was computed over the seeded registry, smaller than the 127-entity population, and no evaluation with real users was conducted.

6 Security, Privacy, and Data Governance

The platform concentrates information about public institutions, their published services, their connection state, and the contact data of their technology officers. That combination is sensitive even when built from public sources, because it enables systematic profiling of institutional weaknesses and because the CIO contact fields constitute personal data. The applicable regime is the personal data protection law [1], together with the transparency and access to public information statute that legitimizes the use of official public sources [2] and the regulatory framework that mandates the interoperability architecture [3]. The design principle is *data minimization by construction*: only the attributes

Table 4. Requirements pending before operational use of the platform.

Requirement	Rationale and required action
Replace simulated connectors	Live national data requires a formal integration agreement and credential management.
Legal instrument for personal data	A documented lawful basis, purpose limitation, retention policy, and database registration are required before processing the CIO contact fields at scale.
Managed secrets and token policy	The signing secret must move to a managed secret store with periodic rotation, and the 120-minute access token must be reviewed against institutional security policy.
Encryption at rest and penetration test	Field-level protection of contact data is recommended, and the applied hardening has not been validated by an external adversarial review.
Availability monitoring and retention	Uptime monitoring and a warm-up strategy are needed for continuous use, together with a formal retention policy for the append-only assessments.

required to compute the indicators are stored, and every value is either institutional data or the professional contact data needed for fieldwork.

Access tokens are signed JWTs (HS256) with a 120-minute access lifetime and a 10 080-minute refresh lifetime; refresh tokens are typed and rejected if presented as access tokens. Passwords are hashed with bcrypt through Passlib. Authorization is role-based: an administrator role is required for every write operation on entities, services, and assessments, while analysts are restricted to reading and analysis. Authentication is hardened against credential stuffing with a lockout rule—five consecutive failures suspend the account for fifteen minutes and return HTTP 423—and every relevant event (successful and failed logins, administrative actions, and access metadata with the client address) is written to an audit table in a non-blocking way, so an audit failure can never break a user transaction. A global middleware enforces a per-address rate limit of 120 requests per minute (HTTP 429 beyond it) and a CORS allowlist restricts which origins may call the API. At the infrastructure level, the database is not published outside the container network, production connections require TLS, and configuration including the signing secret is injected through environment variables rather than committed to the repository. The interface handles token expiry by attempting a refresh and, if it fails, redirecting to authentication.

Two residual risks require explicit attention: the free-tier managed deployment inherits the availability profile of the provider and cannot offer the durability guarantees of an institutional hosting agreement, and the simulated connectors mean that no live data crosses the platform boundary today, so a real AND or MinTIC integration will require an explicit data-processing instrument before any transfer. Table 4 consolidates the requirements pending before operational use.

7 Conclusions and Future Work

Four conclusions follow from the design and the empirical results. First, the automated instrument is technically feasible and produces a coherent diagnosis: operationalizing the four MinTIC domains into explicit indicators combined with a deterministic rubric yields a national index of 2.3/5.0 at the *Basic* level, and the automatic cross-check over the seeded registry reproduces the same domain ordering, which indicates that the result derives from the computation rather than from a predefined narrative. Second, registration is not interoperability readiness: a third of the entities formally enrolled in the central node remain at the initial level, so any policy that measures progress by the number of registered members will overestimate the real state of the ecosystem. Third, the binding constraint is semantic rather than technological, because entities resolve connectivity before they resolve meaning, and investment in integration infrastructure without parallel semantic harmonization does not produce reusable interoperability. Fourth, automated auditing reduces administrative burden while increasing traceability, since every point is attributable to a named indicator and every assessment records its assessor and date, which supports a positive but qualified answer to RQ4 given the limitations of Section 5.2.

Several lines of future work follow. The most urgent is replacing the simulated connectors with authenticated interfaces to the national systems, which requires an institutional agreement and is the first step towards a production-grade diagnostic. Coverage should then be extended to second- and third-level territorial entities, where deficits are expected to be larger, and the semantic layer should be reinforced with concrete artifacts: published data dictionaries, reference schemas, and a conformance test suite the platform can consume automatically. Three research directions are proposed: real-time availability monitoring for critical services, so the diagnostic reflects operational continuity and not only declared availability; predictive analysis of citizen service demand to anticipate which services should be integrated first; and a longitudinal study of the ecosystem's evolution, which the append-only assessment model already supports. Finally, qualitative validation with the technology officers of the assessed entities and a usability evaluation with real analysts would close the gap between an instrument that measures correctly and one that is actually adopted, while comparative studies with other national X-Road implementations would position the Colombian results against international references.

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